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ABSORPTION, SERUM LEVELS AND URINARY EXCRETION OF INORGANIC SULFATE AFTER ORAL ADMINISTRATION OF SODIUM SULFATE IN THE CONSCIOUS RAT

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Summary

The absorption of inorganic sulfate after ingestion was investigated in rats. After oral administration of $\text{Na}_2^{35}\text{SO}_4$, ^{35}S radioactivity was measurable in plasma already after 15 min and its plasma concentration reached a peak after about 1.5–2 h. The ^{35}S -radioactivity excreted in urine during 24 h after ingestion of $\text{Na}_2^{35}\text{SO}_4$ together with varying amounts of unlabelled Na_2SO_4 (0.25–5.0 mmol Na_2SO_4 per rat) indicated an almost complete absorption of inorganic sulfate from the gastrointestinal tract. Determination of the inorganic sulfate concentration in rat serum 2 h after oral administration of 5.0 mmol Na_2SO_4 revealed a three-fold increase in serum sulfate concentration. The data suggest a rapid and almost complete absorption of inorganic sulfate after oral administration in the rat. Its importance in relation to the sulfate availability for sulfate conjugation of drugs is discussed.

Introduction

Many xenobiotics with an appropriate acceptor group are conjugated in mammals with sulfate, by which they generally lose their pharmacological properties and can easily be eliminated in bile or urine. For sulfation, inorganic sulfate is needed, which is converted with ATP to the co-substrate for sulfation, adenosine 3'-phosphate 5'-sulfatophosphate. Inorganic sulfate in blood is immediately available for sulfate conjugation in the liver [1]. Recently it has been shown that at the physiological plasma sulfate concentration the rate of sulfation is not yet at its maximum [2].

It has been suggested that drug that are mainly sulfated in the body can exhaust the directly available sulfate pool when given in high doses to animals and man [3,4]. Under those conditions, orally administered Na_2SO_4 increased the in vivo sulfation, supposedly by compensating the shortage of sulfate [5,6]. In several animal species inorganic sulfate has been shown to be well absorbed after oral administration; however, the available data are incomplete because they have been obtained using very different experimental conditions [7–10].

Surprisingly, many textbooks in pharmacology state that inorganic sulfate is poorly absorbed from the gastrointestinal tract (see, for example, Refs. 11 and 12) and use this postulate to explain the laxative action of Na_2SO_4 and other sulfate salts.

Because of this discrepancy we decided to reinvestigate to what extent orally administered sulfate is absorbed from the gastrointestinal tract in conscious rats. To quantitate accurately the sulfate absorption, ^{35}S -labelled Na_2SO_4 was used, with or without unlabelled Na_2SO_4 . These are the first data on oral sulfate absorption in the rat, giving the rate of absorption, as well as the resulting inorganic sulfate concentrations in serum and the sulfate excretion in urine.

Materials and Methods

Animal experiments

Male Wistar rats (300–330 g body weight) were used for the experiments. They had free access to food and water.

The rate of appearance of [^{35}S]sulfate in blood after its oral administration was studied in rats with a permanent catheter as described by Steffens [13] in the right atrium, which allows the sampling of blood from conscious unrestricted animals. Under light ether anesthesia the animals received 600 $\mu\text{Ci/kg}$ body wt. $\text{Na}_2^{35}\text{SO}_4$ in 2 ml water by gastric tube. When only a tracer dose was given, the 2 ml water also contained 0.3 mmol NaCl (NaCl, 9 g/l); when, in addition to the tracer, unlabelled sulfate was administered, the solution contained 1.0 or 5.0 mmol Na_2SO_4 . Blood was collected in heparinized microtubes (Caraway Micro Blood Collecting Tubes, Sherwood Medical Inc. St. Louis, MO, U.S.A.) at various time intervals, to determine the plasma ^{35}S concentration.

In normal rats, without a permanent catheter, the urinary excretion of sulfate was studied following administration by gastric tube under light ether anaesthesia of $\text{Na}_2^{35}\text{SO}_4$ (50 $\mu\text{Ci/kg}$ body wt.) in 2 ml water with NaCl (0.3 mmol) or with Na_2SO_4 in increasing amounts (0.25 to 5.0 mmol). In these experiments the ^{35}S -radioactivity is concentrated in urine; therefore, a lower dose of radioactivity was given than in the experiments in which ^{35}S -radioactivity was measured in a small sample of plasma. The rats were placed in metabolism cages designed to allow the separate collection of urine and faeces. Urine was collected up to 7 days after oral sulfate administration. Rats receiving the two high sulfate doses (2.5 and 5.0 mmol) were afflicted with diarrhoea, which started about 4 h after sulfate administration and lasted for about 4 h. To prevent contamination of urine by faeces, the watery stools were removed immediately after they were produced. This, however, inevitably resulted in

some loss of urine. As a consequence, at these doses non-contaminated urine samples during the first 8 h were collected.

The serum inorganic sulfate concentration was determined for rats, which received, in 2 ml water, 0.5, 2.5 or 5.0 mmol Na_2SO_4 by gastric tube under light ether anaesthesia. Rats receiving 2 ml water only or with 0.3, 2.5 or 7.5 mmol NaCl added, served as controls. Blood was collected from the aorta under ether anaesthesia 2 h after oral administration. In some groups of rats, serum osmolality, chloride and protein concentrations were also determined.

Materials

$\text{Na}_2^{35}\text{SO}_4$ was obtained from New England Nuclear, Dreieich, F.R.G., as a solution in water (710 Ci/mol). Dextran was of clinical grade (Sigma; average mol.wt. 80 700). Anhydrous Na_2SO_4 and all other chemicals used were of analytical grade.

Analytical procedures

For determination of ^{35}S -radioactivity in plasma or urine, 50 μl samples of plasma or 100 μl samples of urine were added to a scintillation medium (Pico-Fluor 30, Packard Instrument Comp., Downers Grove, IL, USA), and counted for radioactivity in a liquid scintillation spectrometer (Mark I, Nuclear Chicago

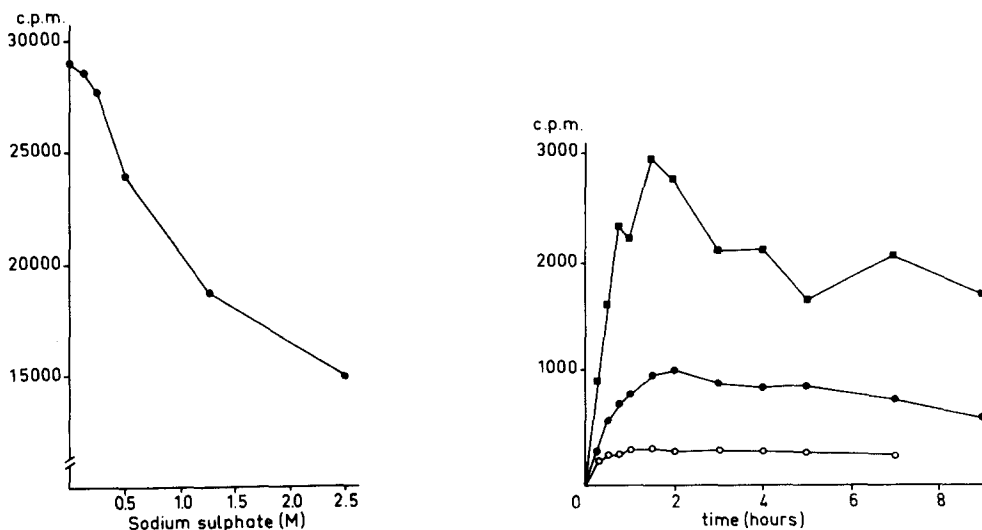


Fig. 1. Decreased efficiency of ^{35}S -radioactivity counting by inorganic sulfate. In each scintillation vial 29 000 cpm of ^{35}S sulfate was present; 10 μl of a solution of unlabelled Na_2SO_4 was added of the concentration indicated in the figure, and scintillation was counted in Pico-Fluor scintillation medium. At increasing inorganic sulfate concentration efficiency decreased, because the sulfate did crystallize in the Pico-Fluor.

Fig. 2. Plasma concentration of ^{35}S radioactivity after oral administration of various doses of $\text{Na}_2^{35}\text{SO}_4$ in the rat. Na_2SO_4 was given orally to rats under ether anaesthesia in a volume of 2 ml; all rats received 600 $\mu\text{Ci/kg}$ body weight $\text{Na}_2^{35}\text{SO}_4$. Blood was sampled via a permanent catheter from the heart. A 50 μl sample of plasma was counted for radioactivity. $\blacksquare$ — $\blacksquare$, typical example of a rat receiving only the tracer dose of radioactivity; $\bullet$ — $\bullet$, mean of two rats receiving tracer dose and 1 mmol of unlabelled Na_2SO_4 ; $\circ$ — $\circ$, mean of three rats receiving tracer dose and 5 mmol of unlabelled Na_2SO_4 .

Inc., Chicago, IL, USA). At increasing sulfate concentration the sulfate crystallized in the Pico-Fluor, which caused a decreased efficiency of counting of the ^{35}S -radioactivity. Fig. 1 shows this effect for increasing concentrations of unlabelled inorganic sulfate. The data are not corrected for this decreased efficiency in radioactivity counting at high sulfate concentration, which occurs in urine at the two highest oral sulfate doses. In a later experiment the correction needed at the highest sulfate dose (5 mmol Na_2SO_4 per rat) was estimated by diluting the urine until the efficiency of counting became constant. A correction of 10–20% was found to be necessary.

The concentration of inorganic sulfate in rat serum was determined with a modification of the turbidimetric method of Berglung and Sörbo [14]. To 0.5 ml of serum 2 ml of trichloroacetic acid (50 g/l) were added and the mixture was allowed to stand for 10 min at room temperature. After centrifugation, 1 ml of the clear supernatant was added to 0.25 ml BaCl_2 reagent (20 g BaCl_2 and 10 g dextran per l water) and the absorbance was read after precisely 35 min at 360 nm against a sample background (1 ml supernatant and 0.25 ml reagent containing 10 g dextran in 1 l water).

Chloride was measured by conductometric titration without precipitation of proteins [15]. The serum protein concentration was determined by the biuret method [16] and the serum osmolality was measured by freezing point depression (Advanced Instr. Osmometer, Advanced Instr. Inc., Newton Island, MA, U.S.A.).

Results

Rate of appearance of [^{35}S]sulfate in plasma after oral administration of $\text{Na}_2^{35}\text{SO}_4$

After oral administration of a tracer dose of $\text{Na}_2^{35}\text{SO}_4$ radioactivity was measurable in plasma after 15 min (Fig. 2). Plasma radioactivity reached a peak level after 1.5–2 h. 10 h after administration, about 50% of the maximum

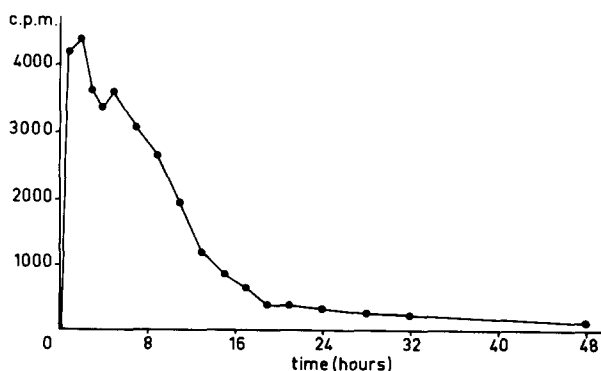


Fig. 3. Appearance of ^{35}S -radioactivity in plasma after oral administration of a tracer dose of [^{35}S]-sulfate to a rat. In 1 ml saline 600 $\mu\text{Ci/kg}$ body wt. of $\text{Na}_2^{35}\text{SO}_4$ was administered orally to a rat under light ether anesthesia. Various times after administration, 50- μl plasma samples were counted for radioactivity.

[^{35}S]sulfate plasma concentration remained, which decreased in 19 h to less than 10% (Fig. 3).

When unlabelled sodium sulfate was added to the administered solution, the peak of radioactivity appeared also 1.5–2 h after the sulfate administration. The amount of radioactivity in plasma decreased as the dose of unlabelled sodium sulfate increased, which indicates that the fractional absorption of sodium sulfate decreases as the administered dose increases.

Excretion of [^{35}S]sulfate in urine after oral administration of Na_2SO_4

When the labelled sodium sulfate (50 $\mu\text{Ci/kg}$ body wt.) is administered in a saline solution or with 0.25 or 0.5 mmol of Na_2SO_4 per rat, about 90% of the administered radioactivity is recovered in urine within 24 h (Table I). In the following days only a small additional amount is excreted, so that after a week a little more than 90% of the ^{35}S -radioactivity is recovered in urine. When the dose of unlabelled Na_2SO_4 was increased to 1.0, 2.5 and 5.0 mmol per rat, the percentage of the dose recovered in urine after 24 h was decreased to 73, 67 and 56%, respectively. However, the recovery data at these higher sulfate doses underestimate the actual excretion because of the urine loss as a consequence of diarrhoea and because of the decreasing efficiency of ^{35}S -radioactivity counting caused by high concentration of inorganic sulfate in the urine sample. Since we did not measure the sulfate concentration in urine, it was not possible to correct for the decreased efficiency of counting. In order to determine the size of this correction, we have administered 5 mmol labelled sulfate to an additional group of rats ($n = 4$) and determined the 24 h urinary excretion of ^{35}S -radioactivity. Without correction we recovered $41.4 \pm 5.1\%$ of the dose (mean $\pm$ S.E.), but after dilution of urine to maximal efficiency of counting we determined a recovery of $47.3 \pm 5.1\%$. This shows that the high concentration of sulfate in urine does decrease the recovery by about 15%.

Serum sulfate concentrations 2 h after oral administration of inorganic sulfate

Our experiments with ^{35}S -labelled sulfate indicated that about 2 h after oral sulfate administration, a peak in plasma sulfate concentration was reached.

TABLE I

URINARY EXCRETION OF VARIOUS ORAL DOSES OF $\text{Na}_2^{35}\text{SO}_4$

The percentage of the dose of ^{35}S radioactivity recovered in urine after various time intervals is given for different doses of Na_2SO_4 (mmol per rat). In all groups of rats the same amount of [^{35}S]sulfate was injected (50 $\mu\text{Ci/kg}$). Values are the mean $\pm$ S.E. for four rats per group.

Dose of Na_2SO_4 (mmol/rat)	Percentage of ^{35}S radioactivity in urine after:			
	8 h	24 h	48 h	7 days
0.0	—	84.0 ± 2.2	88.1 ± 2.1	91.0 ± 2.1
0.25	—	88.9 ± 2.9	91.7 ± 2.5	93.2 ± 2.6
0.5	—	89.7 ± 0.9	92.3 ± 1.2	93.9 ± 1.3
1.0	—	72.9 ± 2.8	78.1 ± 4.0	79.7 ± 4.2
2.5	39.6 ± 1.5	65.8 ± 2.1	68.5 ± 2.4	69.4 ± 2.4
5.0	29.6 ± 2.0	53.6 ± 3.6	56.5 ± 3.7	57.4 ± 3.7

TABLE II

EFFECT OF ORALLY ADMINISTERED Na_2SO_4 AND NaCl ON SERUM SULFATE CONCENTRATION IN THE RAT

Serum sulfate concentration was determined 2 h after oral administration of various doses of Na_2SO_4 and NaCl , dissolved in 2 ml water. Blood was collected from the aorta under ether anaesthesia and inorganic sulfate was determined turbidimetrically. All results are the mean $\pm$ S.E. for six rats per group.

Orally administered in 2 ml per rat	Serum sulfate concentration (mmol/l)
Controls	0.77 ± 0.04
Na_2SO_4 0.5 mmol	$1.34 \pm 0.06^*$
2.5 mmol	$1.95 \pm 0.15^*$
5.0 mmol	$2.13 \pm 0.09^*$
NaCl 0.3 mmol	$0.66 \pm 0.03^{**}$
2.5 mmol	$0.57 \pm 0.02^*$
7.5 mmol	$0.58 \pm 0.02^*$

* Significantly different from control (nothing administered) at $P < 0.005$ (Wilcoxon's test).

** Significantly different from control (nothing administered) at $P < 0.05$ (Wilcoxon's test).

Therefore we determined in another series of experiments the serum concentration of inorganic sulfate 2 h after oral administration of unlabelled sodium sulfate (Table II). The results we obtained in fed rats were not significantly different from those obtained in rats which had fasted 24 h before the experiment (data not shown).

A dose of 0.5 mmol Na_2SO_4 caused a significant increase in serum sulfate concentration compared with the control situation (nothing administered). Raising the sulfate dose to 2.5 or 5.0 mmol, Na_2SO_4 increased serum sulfate even more. Because the sulfate solutions administered were strongly hypertonic, NaCl solutions with a comparable osmolality were given as a control. We found that, already at an oral dose of 0.3 mmol NaCl in 2 ml water, NaCl decreased the serum sulfate concentration (Table II). Oral administration of 2.5 or 7.5 mmol NaCl caused, both to the same extent, a more profound decrease in the serum sulfate concentration, whereas oral administration of water had no influence on the serum sulfate concentration.

To obtain more information about possible shifts of body water to the gastrointestinal tract as a consequence of the administration of a hypertonic solution, we determined osmolality, chloride concentration and total protein in serum, along with inorganic sulfate (Table III). Oral administration of 2 ml distilled water had no influence on the sulfate concentration, chloride concentration and osmolality in serum as compared with values from control rats, while they were elevated by 5 mmol of sodium sulfate. An NaCl dose of the same osmolality as the highest Na_2SO_4 dose caused increases in serum osmolality and chloride, but a decrease in serum sulfate. Total protein did not differ significantly from control in either group.

After blood sampling, we examined the state of the stomach and intestine. After administration of the hypertonic solutions of NaCl and Na_2SO_4 the wall of the stomach and the upper part of the small intestine was swollen, and the lumen contained more liquid after Na_2SO_4 administration than after NaCl

TABLE III

EFFECT OF ISOMOLAR DOSES OF Na_2SO_4 AND NaCl ON SERUM SULFATE, CHLORIDE, PROTEIN AND OSMOLALITY

Serum was obtained 2 h after oral administration of distilled water, Na_2SO_4 or NaCl (2 ml of each solution); controls received nothing, but were only briefly anesthetized with ether as were the other rats for oral administration of the solutions. See further the legend to Table II. Each group consisted of six rats.

Orally administered (in 2 ml/rat)	Sulfate concn. (mmol/l)	Osmolality (mosmol/kg)	Chloride concn. (mmol/l)	Protein concn. (mg/l)
Controls	0.76 ± 0.04	296.3 ± 0.7	104.2 ± 0.5	64.4 ± 1.1
Water	0.79 ± 0.01	296.2 ± 0.7	103.8 ± 0.6	63.6 ± 0.7
Na_2SO_4 (5 mmol)	$2.42 \pm 0.08^*$	$302.0 \pm 0.3^*$	$106.3 \pm 0.4^{**}$	65.8 ± 0.7
NaCl (7.5 mmol)	$0.56 \pm 0.01^*$	$303.5 \pm 1.1^*$	$111.3 \pm 0.5^*$	62.8 ± 0.9

* Significantly different from controls and from rats receiving water, $P < 0.005$.

** Significantly different from controls and from rats receiving water, $P < 0.01$.

administration. In addition, Na_2SO_4 caused hyperaemia of the small intestine. 2 ml distilled water had no visible effect on stomach or intestine.

Discussion

The absorption of orally administered Na_2SO_4 in freely moving rats is very rapid. ^{35}S -radioactivity was detected in plasma soon after administration. When a tracer dose was given, a peak in radioactivity was reached after 1.5–2 h. With larger amounts of Na_2SO_4 the level of ^{35}S -radioactivity remained for longer near the peak level. When 5 mmol of Na_2SO_4 was given, a steady state was almost reached between the absorption from the gastrointestinal tract and the elimination from the blood.

From these experiments no information was obtained about the quantities of sulfate absorbed. This information came from the results of the experiments on urinary excretion of orally administered $\text{Na}_2^{35}\text{SO}_4$ together with increasing amounts of unlabelled sulfate. At the lower doses about 90% of the radioactivity was excreted in urine, indicating almost complete absorption of the dose from the gut. The loss of about 10% of the total [^{35}S]sulfate dose can in part be explained by the incorporation of ^{35}S into macromolecules in the body. Although the percentage recovered in urine decreased at higher sulfate doses, the absolute amount of sulfate absorbed strongly increases. Since the highest doses of sulfate caused diarrhoea it is to be expected that sulfate is lost in the faeces and recovery in urine must be lower because less time is available for absorption.

From the data in Table I it is obvious that in the case of the highest sulfate dose, half of the total radioactivity excreted in urine in 24 h had already been excreted after 8 h. After 24 h little additional ^{35}S -radioactivity was excreted. This emphasizes the rapidity of sulfate absorption.

As for the high urinary recovery of orally administered [^{35}S]sulfate, the results of Berry et al. [8] point in the same direction. They found in swine, after oral administration of sodium [^{35}S]sulfate, a peak level of radioactivity in serum after 3 h. After 7 days 58% of the dose was excreted in urine against

only 5% in the faeces. A high recovery of oral sulfate was also found by Dziewiatkowski [7] in the rat and more recently by O'Connor et al. [9] in the dog.

In order to evaluate the increase in the serum sulfate concentration and the possible concomitant water shifts, we have measured osmolality, inorganic sulfate, chloride and protein concentration in serum after oral Na_2SO_4 administration (Table III). If we assume the distribution volume for chloride and sulfate ions to be 33% and 25% of body weight, respectively [17,18], and total distribution volume of osmotically active particles (total body water) to be 67% of body weight [18], the following calculation can be made for the rats of 300 g used in these experiments. The increase of the mean serum sulfate concentration from 0.8 to 2.4 mmol/l after sulfate administration (Table III) corresponds to the addition of 0.12 mmol sulfate to the sulfate distribution volume, which increases osmolality of the body water by 1.8 mosmol/kg. The increase of chloride in plasma from 104 to 106 mmol/l most likely is due to a shift of body water to the gastrointestinal tract. It can be calculated that this shift amounts to 2.4 ml water, which raises the osmolality of serum with 3.6 mosmol/kg. Thus, sulfate absorption and water movement together can account for the observed change in osmolality of serum from 296.2 to 302.0 mosmol/kg.

The increase of osmolality after oral administration of NaCl (Table III) can be completely accounted for by the increased serum chloride concentration, which implies that, on the moment of blood sampling, NaCl had not caused a net water movement from the body fluids to the gastrointestinal tract. It is remarkable that oral administration of NaCl, even physiological saline decreased serum sulfate. Further investigations are required to solve the underlying mechanism.

Clearly the laxative action of sulfate salts cannot be due to their lack of absorption in the intestine. An alternative explanation of this action is that the high sulfate concentration irritates the intestinal mucosa, as is suggested by the hyperemic state of intestine which we observed, which causes an accelerated transit of intestinal contents. A rapid accumulation of water in the intestinal lumen by osmotic action may add to the laxative effect.

The toxicological implication of these findings are obvious: since many toxic compounds are detoxified by sulfate conjugation, the availability of sulfate may become decisive in this process. Oral supplementation of sulfate under conditions of sulfate depletion at high doses of toxicants clearly relieves this. On the other hand, some compounds containing an *N*-hydroxy group become more toxic by sulfation [19]. High inorganic sulfate in the food therefore might enhance the toxicity of this type of compound, since (in the rat) sulfation is not yet saturated in respect to inorganic sulfate at physiological plasma sulfate levels [2].

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